

¹³N - Comments on evaluation of decay data

by X. Mougeot

*Université Paris-Saclay, CEA, List, Laboratoire National Henri Becquerel (LNE-LNHB),
91120 Palaiseau, France*

This evaluation has been completed with the literature available by February 2026. The former evaluation was completed in 2002.

This radionuclide is one of the $T_z=1/2$ mirror β transitions surveyed in order to test the predictions of the Standard Model, and to search for possible signals of new physics [2023Se01].

1 Nuclear data

The ground state of ¹³N decays at 100% by EC/ β^+ emission to the ground state of ¹³C. Indeed, the energy of the first excited state of ¹³C is 3089.451 (19) keV [2024Ke04], above the adopted Q-value of 2220.47 (27) keV from AME2020 [2021Wa16].

The spins and parities of these two ground states have been adopted from [2024Ke04]: both are $1/2^-$. The ground state of ¹³C is stable.

1.1. Half-life

Table 1 summarizes the measured half-lives of ¹³N decay available in the literature. The whole dataset is discrepant ($\chi^2 = 2.02$ and $\chi^2_{\text{crit.}} = 1.96$). The values from [1945Sieg, 1948Co24, 1961Ra06] are excluded by Chauvenet criterion. The dataset is discrepant due to the two most precise measurements from [1960Ja12, 2022Lo14], which are consistent ($\chi^2 = 5.72$ and $\chi^2_{\text{crit.}} = 6.63$) but in tension at the 2.4σ level. Their weighted average with internal uncertainty is 9.9549 (27) min. The authors in [2022Lo14] proposed rejecting all the measurements realized prior to 1969 arguing that analysis via the χ^2 method can introduce bias and the maximum-likelihood is preferable. Eventually, they considered only the values from [1968Ri15, 1977Az01, 2022Lo14]. However, the analyses of the two oldest values were performed with the χ^2 method. The analysis in [2022Lo14] to establish a reference ¹³N half-life of 9.9532 (37) min has not been considered as sufficiently robust.

In the present evaluation, the whole dataset has been considered in order to mitigate the tension between the values from [1960Ja12, 2022Lo14]. The uncertainty of the [1960Ja12] value has been increased from 0.005 min to 0.006 min, which is sufficient to make the dataset consistent ($\chi^2 = 1.95$ and $\chi^2_{\text{crit.}} = 1.96$). The adopted value of the ¹³N half-life corresponds to the weighted average with external uncertainty, i.e., 9.9589 (37) min. The weight of the value from [2022Lo14] has been limited to 50%.

1.2. Beta transition

The ground-state-to-ground-state EC/ β^+ transition is of allowed nature. The BetaShape program [2019Mo35, 2023Mo21] has been used to determine the average energy of the continuous spectrum, the fractional atomic shell electron-capture probabilities, the EC/ β^+ ratios and the log ft value. The EC/ β^+ splitting of the branch has been deduced as $P_{0,0}(\text{EC}) = 0.197$ (5) % and $P_{0,0}(\beta^+) = 99.803$ (5) %.

The average energy of the β^+ spectrum has been calculated as $E_{av} = 492.21$ (12) keV from the experimental shape factor $C(W) = 1 + 0.0014W$ from [1958Da09]. $P_{0,0}(EC)$ and $P_{0,0}(\beta^+)$ have been determined from the EC/β^+ ratio of 0.00197 (5). This is in excellent agreement with the $P_{0,0}(EC)$ value of 0.196% given in [2023Se01], which was calculated from tabulated values in [1977Ba48]. The K/β^+ ratio from BetaShape is 0.00182 (5), in good agreement with the measured value from [1963Le22], $K/\beta^+ = 0.00168$ (12). The $\log ft$ value has been determined to be 3.67176 (44), in excellent agreement with the calculated value of 3.67181 (44) from [2023Se01] when including the radiative correction.

Table 1 – Measured half-lives of ^{13}N decay.

Reference	$T_{1/2}$ (min)	Comment
1935Ru01	14	No uncertainty; excluded.
1939Wa09	9.930 (30)	
1945Sieg	10.13 (10)	Excluded by Chauvenet criterion.
1948Co24	10.20 (10)	Excluded by Chauvenet criterion.
1950Ho01	10.05 (10)	
1953Ch34	10.048 (32)	
1955Wi43	10.080 (40)	
1957Da08	9.960 (30)	
1957De22	10.02 (10)	
1957No17	10.07 (6)	
1958Ar15	9.960 (30)	
1960Ja12	9.965 (6)	Original uncertainty increased from 0.005 to 0.006.
1960Ki02	9.93 (5)	
1961Ra06	12.3 (7)	Excluded by Chauvenet criterion.
1965Bo42	10.05 (5)	
1965Eb01	9.960 (20)	
1968Ri15	9.963 (9)	
1973SiYS	10.0 (5)	
1977Az01	9.965 (10)	
1980An40	9.967 (10)	
1989KaYR	9.962 (20)	
2022Lo14	9.9508 (32)	
Adopted	9.9589 (37)	Weighted average with external uncertainty

The measurements of the maximum energy of the β^+ continuous spectrum reported in the literature are summarized in Table 2. The dataset is discrepant ($\chi^2 = 6.44$ and $\chi^2_{crit.} = 2.80$) due to the values from [1941To01, 1957Da08] whose uncertainties seem underestimated. Therefore, they have been reasonably increased from 4 to 10 keV for [1941To01] and from 3 to 5 keV for [1957Da08]. The resulting dataset is consistent ($\chi^2 = 2.44$ and $\chi^2_{crit.} = 2.80$). The evaluated maximum energy corresponds to the weighted average with internal uncertainty: $E_{max} = 1199.5$ (29) keV. It is consistent with the adopted maximum energy of 1198.47 (27) keV deduced from the Q-value [2021Wa16].

Table 2 – Measured β^+ endpoint energy of ^{13}N decay.

Reference	$E_{\max}$ (keV)	Comment
1939Ly15	1198 (6)	
1941To01	1218 (4)	Uncertainty increased from 4 to 10 keV.
1945Sieg	1240 (20)	
1948Co24	1250 (30)	
1950Ho01	1202 (5)	
1954Gr66	1185 (25)	
1957Da08	1190 (5)	Uncertainty increased from 3 to 5 keV.
1958Da09	1190 (3)	Same authors and study as [1957Da08]; excluded.
Evaluated	1199.5 (29)	Weighted mean with internal uncertainty
Adopted	1198.47 (27)	From AME2020 [2021Wa16]

2 Atomic data

The fluorescence yield ω_K has been taken from Schönfeld *et al.* [1996Sc06].

3 Gamma emission

The annihilation radiation emission probability, without the correction factor for the annihilation-in-flight process in the medium, has been calculated by Saisinuc [2008DuZX] from the related decay data as $(2 \times P_{\beta^+})$: $I_{\gamma^\pm} = 199.606$ (10) %.

4 References

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